

JOSEPH D VIVIANO

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EXPERIENCE

ML Scientist

2022–Now

[Yoshua Bengio's Group](#), [Mila Québec AI Institute](#)

- Lead developer of [TorchGFN](#), a modular and easy to extend open source GFlowNet library (an amortized sampling method used in many AI for science applications).
- Collaborations with [Amgen](#) (ML for antibody design) and [Intel](#) (large scale training of GFlowNets).
- Developed a collaboration and wrote a grant proposal with the [Joint BioEnergy Institute](#) to develop a multi-center AI for Metabolic Engineering research collaboration.
- ML engineering support for the lab, largely to optimize code-bases for efficient training (e.g., [RGFN](#) and [FLECS](#).)

ML Research Scientist

2021–22

[Deep Genomics](#)

- I investigated architectures for sequence-based RNA foundation models, methods for cell-type generalization (a core challenge for drug discovery), & anomaly detection pipeline for the drug screening platform allowing for early detection of laboratory failures.

Scientific Mentor

2021–Now

[Creative Destruction Lab Montréal](#)

- Performed technical assessment & consulting for early-stage startups.
- Guidance on ML and AI strategy towards next funding round.

Applied Research Scientist

2020–21

[Mila Technology Transfer](#), [Mila Québec AI Institute](#)

- Collaborative development of a deep-learning driven equity momentum trading strategy ([CDPQ](#)) & digital forgery detection ([Jumio](#)).
- Research on LLM data set composition & modular deep learning.

Fellow

2021

[PhD to VC, Fifty Years](#)

- I sourced and performed due-diligence on pre-seed/seed deep-tech companies, & helped source technical talent.

Research Intern

2020

[Google](#), [Ad Predicted Click Through Rate Team](#)

- Uncertainty estimation research for predicted click through rates.

Research Intern

2019

[Imagia Cybernetics](#)

- Cancer localization methods (segmentation-free localization & weak supervision from clinical notes) addressing scarcity of labelled data.

Research Methods Specialist

2014–17

[Kimel TIGRlab](#), [Centre for Addiction and Mental Health](#)

- Biomarkers for vulnerable [schizophrenia](#) and [Alzheimer's](#) patients.
- Designed, built, and managed (team of 5) a [data management platform](#) 22-node compute cluster, and [QA tools](#) used by team of 20.

PROJECTS

TorchGFN 259 stars, 49 forks, 13 contributors. A modular, easy to extend GFlowNet library. [JMLR 2026 \(Accepted\)](#)

TorchXRayVision 995 stars, 227 forks, 11 contributors. Open source library for XRay datasets & models. [PMLR 2022](#)

EDUCATION

MSc. CS, Machine Learning

2018–20

[Mila](#), Université de Montréal, Montréal, QC

Research on utility of saliency methods for [diagnosing generalization failures](#) of deep learning models.

MSc. Bio, Neuroscience, Cum Laude

2011–13

[Schneider Lab](#), York University, Toronto, ON

Subcortical fMRI methods (1, 2).

BSc. Psychology, Hons.

2005–09

Queen's University, Toronto, ON

INVITED TALKS

FACSS SciX Keynote Presentation

2024

Keynote presentation on AI for science.

JGI Annual User Meeting

2023

Considerations for Machine Learning-Aided Lab in the Loop Scientific Discovery.

RESEARCH & COMMUNITY

• **Teaching:** Stats & ML taught at [Center for Addiction and Mental Health](#) & [Montréal AI & Neuroscience Meeting, 2019](#).

• **Paper Reviews:** [NeurIPS](#), [ICLR](#), [Biological Psychiatry](#), [MIDL](#), & [PLOS ONE](#).

• **Area Chair:** [NeurIPS Datasets & Benchmark Track, 2023–2024](#)

PUBLICATIONS

* denotes first/last author. h-index=20

• Action abstractions for amortized sampling, [ICLR 2025](#).

• * What's in the Box? A Preliminary Analysis of Undesirable Content in the Common Crawl Corpus, [ACL 2021](#).

• * Saliency is a Possible Red Herring When Diagnosing Poor Generalization, [ICLR 2021](#).

• * Resting-state connectivity biomarkers of cognitive performance and social function in individuals with schizophrenia spectrum disorder and healthy control subjects, [Biological Psychiatry 2018](#).